

My research focuses on how the properties of the interstellar medium (ISM), including gas and dust grains, vary across environments that differ from the local ISM of the Milky Way. Fundamental studies of the ISM have been focused on the thin-disk region within ~ 2 kpc of the Solar System and ~ 0.1 kpc of the Galactic plane (1–3); consequently, the Galactic ISM beyond this region remains much less well understood and difficult to observe (4). Even less known about the properties of gas and dust in the ISM of distant galaxies (5). I aim to (A) investigate dust extinction across nearby and distant galaxies; (B) connect dust extinction, dust depletion, and gas physical conditions into a consistent physical picture; (C) characterize diversity in dust and gas properties in the Milky Way beyond the area of local ISM studies.

The central question is whether our understanding of the gas and dust physics in the local ISM can be broadly applied to studies of the ISM in other galaxies? Beyond the fundamental question of the Milky Way’s place among galaxies, answering this question has implications for extragalactic astrophysics and cosmology. For example, corrections for Galactic and distant galaxies extinction/attenuation affect the interpretation of galaxy observations, including inferred luminosities, colors, stellar masses, star-formation rates, stellar-population ages, metallicities, and distances (6).

Measuring ISM properties in distant galaxies with methods consistent with those used for Milky Way studies is challenging. It requires carefully selected targets and spectroscopy from the largest ground-based observatories (VLT and Keck) and space-based telescopes (HST and JWST). Over the next 5 years, progress will come from large optical and IR surveys with the Vera C. Rubin Observatory and the Nancy Grace Roman Space Telescope. Over the next decade, 30-m-class telescopes such as the ELT and GMT will greatly improve the sensitivity of ground-based observations, while the PRIMA mission (7) will open new opportunities for mid- and far-IR astronomy. At the same time, this progress will require improved physical modeling and more robust interpretation of ISM observations.

Background: My expertise lies in spectroscopic studies of gas and dust absorption and extinction in the ISM of nearby and distant galaxies, using the spectra of background quasars, and modeling the physical state of the ISM and inferring its physical conditions.

During my Ph.D., I studied molecular gas (H_2 , HD, and CO) in the ISM of distant galaxies at cosmic noon ($z \sim 2\text{--}3$) (8–13). Molecular gas has a small cross section ($1\text{--}10$ pc) and is therefore rarely detected toward background quasars (fewer than 1% of sightlines). Moreover, molecular absorption lines are very narrow ($\sim 1\text{--}10$ km s $^{-1}$), so their detection and analysis require high-resolution spectra ($R = \lambda/\Delta\lambda \sim 50,000$) obtained with large telescopes. In (8), I and Dr. S. Balashev developed a method that uses the statistical power of the regular structure of H_2 absorption lines to search for strong molecular H_2 absorbers in low-resolution quasar spectra from SDSS. We were awarded VLT time to confirm the candidates (12) and discover the most saturated known H_2 absorber, likely probing the ISM of a galaxy disk at $z = 2.78$ (13).

Molecules are particularly valuable because they provide precise *diagnostics of the physical conditions* in the ISM. The populations of molecular rotational levels (H_2 , HD, CO) and fine-structure levels of neutral C (C I) are sensitive to collisional excitation and indirect excitation through UV pumping. I have developed a method that compares observed excitation of these species with the photon-dominated region (PDR) simulations (14) to infer gas density, kinetic temperature, and UV radiation-field intensity (12, 15, 16) [Fig.1a]. We found that molecular gas in the Magellanic Clouds (LMC and SMC) is denser and exposed to stronger UV radiation than in the Galactic ISM, while distant galaxies show even more extreme physical conditions (16, 17) [Fig.1b]. Additionally, using CO and C I absorbers, we obtained the most precise constraints on the CMB temperature at $z=1.7\text{--}3$ (18).

Although quasar spectroscopy provides precise measurements of the ISM along the sightlines, information about the host galaxies of the absorbers is often unavailable. In collaboration with Dr. Kulkarni and Dr. Shaban, I search for host-galaxy emission in archival integral-field spectroscopy data for distant quasars with strong absorbers using the VLT, Keck, Gemini, and JWST (19). Independently, I lead the analysis of the HST COS-MaNGA survey (20, 21) that uses UV absorption

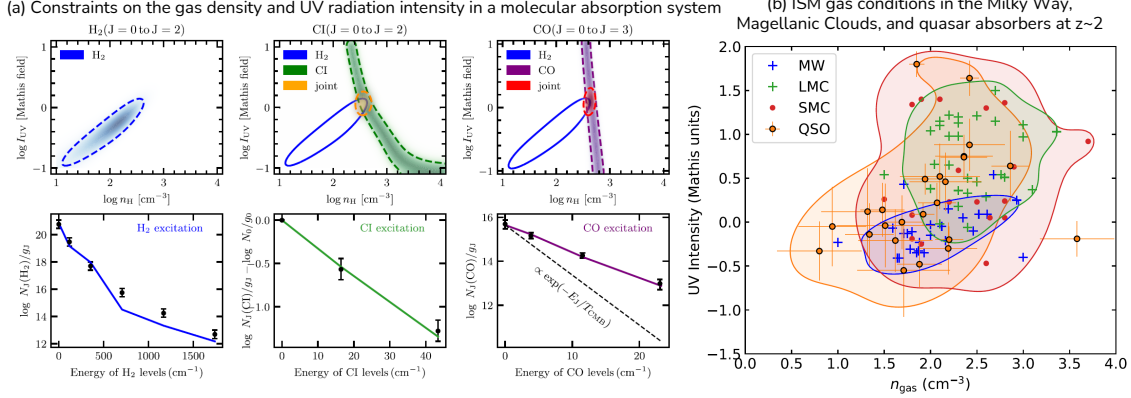


Figure 1: Left: Modeling physical conditions in a molecular cloud using the excitation of H_2 , CO rotational, and C I fine-structure levels (15, 16); Middle: Comparison of constraints on the gas density and interstellar UV radiation intensity in the Milky Way (blue), LMC (green), SMC (red), and quasar absorption systems at $z = 1.5 - 3$ (orange) (16, 17).

spectroscopy to investigate neutral gas and its distribution around nearby SDSS MaNGA galaxies ($z < 0.1$) with detailed IFS maps, with the goal of better understanding the connection between galaxies and their surrounding gas.

On the other hand, cosmic dust is an essential component of the ISM that can also be studied in distant galaxies through the spectra of background quasars. Unlike gas absorption, dust grains scatter and absorb quasar light, resulting in reddened quasar spectra and broad, shallow absorption features, primarily the 2175 Å feature in the UV (tracing small carbonaceous grains, $a < 0.01 \mu\text{m}$) and the 10 and 18 μm silicate features in the mid-IR (tracing larger silicate grains) [Fig. 2a-c]. I work on ongoing JWST programs using MIRI/MRS in Cycles 1 and 3. For Cycle 1 data, we found that dust in galaxies at $z=0.5-1$ may be more silicate-rich from dust in local ISM (22, 23) and has a distinct grain-size distribution [Fig. 3a-c]. Using JWST Cycle 3 data, we have, for the first time, detected ice absorption from dense ISM in normal galaxies at $z \sim 0.7$ (24).

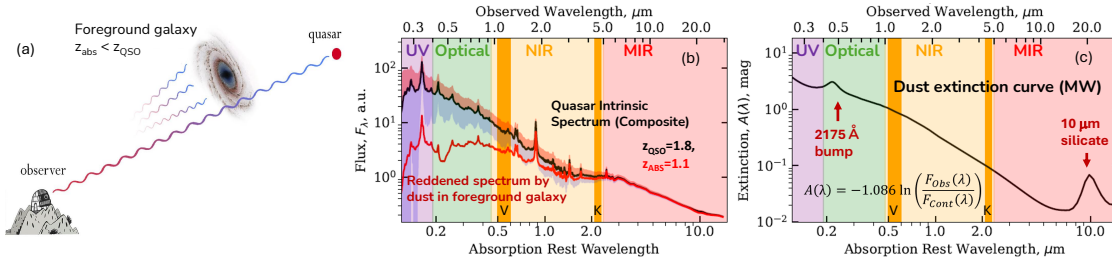


Figure 2: (a) Sketch of the extinction studies in intervening galaxies with a quasar spectroscopy. (b) Reddening in a quasar spectrum ($z=1.8$) by dust in a foreground galaxy at $z=1.1$. (c) Dust extinction curve across UV to IR and strong dust features - 2175 Å bump and 10 μm silicate.

Research plans for the next 5-10 years: I propose three projects that combine studies of dust and gas in the ISM of the Milky Way and nearby and distant galaxies to understand its physical nature and evolution.

(A) Multi-wavelength survey of extragalactic dust extinction.

Previous studies of dust extinction in optical quasar spectra (SDSS) have largely focused on detecting the broad 2175 Å feature associated with absorption systems at $z \sim 1 - 3$ (e.g., 29). Because the intrinsic luminosities of quasars are unknown, these studies could not measure absolute extinction curves and robustly constrain extinction parameters such as A_V and R_V . However, these measurements become possible with IR observations, where the reddening is weaker.

Preliminary studies using SDSS and IR photometric data (2MASS, UKIRT, and WISE) show significant variations in extinction curves relative to the average Milky Way extinction curve (23) [Fig. 3a,b]. I have been awarded time on HST, Keck, and the VLT to conduct a UV-optical-NIR

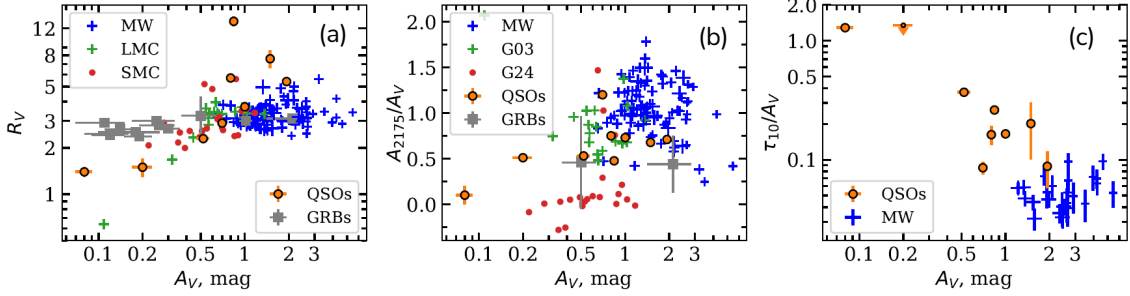


Figure 3: Extinction parameters (A_V and R_V (a)), strength of 2175 Å UV bump (A_{2175} , (b)) and silicate optical depth (τ_{10} , (c)) measured in the MW, SMC, LMC (25–27) and quasar’s absorbers (23) and GRBs at $z \sim 1$ (28). Dust in distant galaxies along quasar sightlines (in orange) appears silicate-rich and has UV-optical extinction distinct from those in local galaxies.

extinction survey of ~ 30 quasar absorption systems at $z \sim 0.5 - 3$, and have submitted several related JWST proposals. To improve multiwavelength extinction-curve analyses of quasar and GRB-afterglow spectra, I am collaborating with Dr. Tayyaba Zafar to incorporate physical dust models with varied grain compositions and size distributions.

Over the next 5 years, I plan to use data from the VLT 4MOST survey (collaborators: Celine Peroux, Annalisa De Cia, Tayyaba Zafar) and the Roman High Latitude Wide Area Survey, which will provide grism spectroscopy for thousands of quasars with strong foreground absorbers. By combining these spectra with deep imaging from the Roman and Rubin surveys, I will establish the relationship between extinction and the host galaxies of the absorbers. Over the next decade, the PRIMA mission will provide far-infrared spectroscopy, enabling studies of silicate dust in galaxies at $z > 1$ and extinction in galaxies at $z \sim 5$, about 1 billion years after the Big Bang.

(B) Dust extinction, dust depletion and gas physical conditions.

About $\sim 50 - 80\%$ of dust grains grow in the ISM (30). Therefore, we may expect that grains composition and size-distribution should be linked to the ISM conditions, including metallicity, depletion, density, temperature, UV radiation, and ionization rate. By combining extinction-survey data with gas-absorption studies, I aim to investigate this relationship. Archival FUSE and HST observations provide absorption-line data for ~ 100 of Milky Way sightlines with measured extinction curves, for ~ 30 sightlines in the Magellanic Clouds, and about ~ 20 sightlines in quasar optical spectra. For the analysis of physical conditions, I will use the methods developed in (16) [Fig.1a]. This sample provides a strong database for testing correlations in the Milky Way ISM and investigating their variations in other galaxies.

For the quasars sample discussed in (A), I will investigate the relationship between extinction and metal depletion and compare it with Galactic studies (e.g., 31). I am collaborating on this work with Dr. Annalisa De Cia, who has measured gas-phase metal depletion in quasar absorbers (32).

(C) Variations of dust and gas properties across the Milky Way environments.

While dust and gas have been extensively studied within ~ 2 kpc of the Solar System in the thin Galactic disk, much less is known about their properties above the disk, in the thick disk and halo. This gas is more diffuse and highly ionized, likely also altering the balance between small and large dust grains (e.g., 4). This region is particularly important because all extragalactic observations are affected by dust extinction in Galactic halo. Similarly, many quasar sightlines with large impact parameters from foreground galaxies likely probe halo gas rather than the ISM, and may therefore exhibit dust properties different from those of Galactic disk dust (23) [Fig. 3a-c]. This project will analyze archival IUE, FUSE, HST, Spitzer, and JWST observations of $\sim 1,000$ sightlines toward distant O- and B-type stars. In collaboration with Dr. Karl Gordon, we will model the stellar atmospheres and correct for disk extinction to isolate the weak signal from extraplanar dust.

These projects (A–C) will be supported through the NASA Astrophysics Data Analysis Program (ADAP) and observatory data-analysis funding programs (e.g., HST, JWST and Roman). **They will provide opportunities for undergraduate students to participate in research, including data analysis, programming, and interpretation of astronomical observations.**

References

- 1* E. L. Fitzpatrick *et al.*, *ApJ* **663**, 320–341 (2007). *2* Y. Sheffer *et al.*, *ApJ* **687**, 1075–1106 (2008). *3* E. B. Jenkins, *ApJ* **700**, 1299–1348 (2009). *4* J. M. Shull *et al.*, *ApJ* **961**, 204 (2024). *5* R. A. Jorgenson *et al.*, *ApJ* **722**, 460–490 (2010). *6* C. Conroy, *ARA&A* **51**, 393–455 (2013). *7* J. Glenn *et al.*, *Journal of Astronomical Telescopes, Instruments, and Systems* **11**, 031628 (2025). *8* S. A. Balashev *et al.*, *MNRAS* **440**, 225–239 (2014). *9* V. V. Klimenko *et al.*, *MNRAS* **448**, 280–298 (2015). *10* S. A. Balashev *et al.*, *A&A* **575**, L8 (2015). *11* V. V. Klimenko *et al.*, *Astronomy Letters* **42**, 137–162 (2016). *12* S. A. Balashev *et al.*, *MNRAS* **490**, 2668–2678 (2019). *13* S. A. Balashev *et al.*, *MNRAS* **470**, 2890–2910 (2017). *14* F. Le Petit *et al.*, *ApJS* **164**, 506–529 (2006). *15* V. V. Klimenko *et al.*, *MNRAS* **498**, 1531–1549 (2020). *16* V. V. Klimenko *et al.*, *MNRAS* **533**, 1367–1393 (2024). *17* D. N. Kosenko *et al.*, *MNRAS* **528**, 5065–5079 (2024). *18* V. V. Klimenko *et al.*, *MNRAS* **498**, 1531–1549 (2020). *19* A. Shaban *et al.*, “Survey of DLA Host Galaxies at $z \sim 2$ –3 in the KCWI Archive”, in preparation, 2026. *20* V. V. Klimenko *et al.*, *ApJ* **954**, 115 (2023). *21* V. Klimenko *et al.*, “The COS-MaNGA project. Analysis of the CGM around SDSS Manga galaxies.”, in preparation, 2026. *22* V. V. Klimenko *et al.*, *ApJ* **998**, 299 (2026). *23* V. V. Klimenko *et al.*, *ApJ* **1003**, 129 (2026). *24* V. Klimenko *et al.*, “Mining for Diamonds in a Lensed Galaxy ~ 7 Billion Years Ago”, submitted to Nature, 2026. *25* K. D. Gordon *et al.*, *ApJ* **594**, 279–293 (2003). *26* K. D. Gordon *et al.*, *ApJ* **705**, 1320–1335 (2009). *27* K. D. Gordon *et al.*, *ApJ* **970**, 51 (2024). *28* T. Zafar *et al.*, *MNRAS* **479**, 1542–1554 (2018). *29* J. Ma *et al.*, *MNRAS* **472**, 2196–2220 (2017). *30* B. T. Draine, *Physics of the Interstellar and Intergalactic Medium* (Princeton University Press, 2011). *31* W. Zuo *et al.*, *ApJS* **257**, 63 (2021). *32* C. Konstantopoulou *et al.*, *A&A* **681**, A64 (2024).